

Ch 8. Path Integral Control

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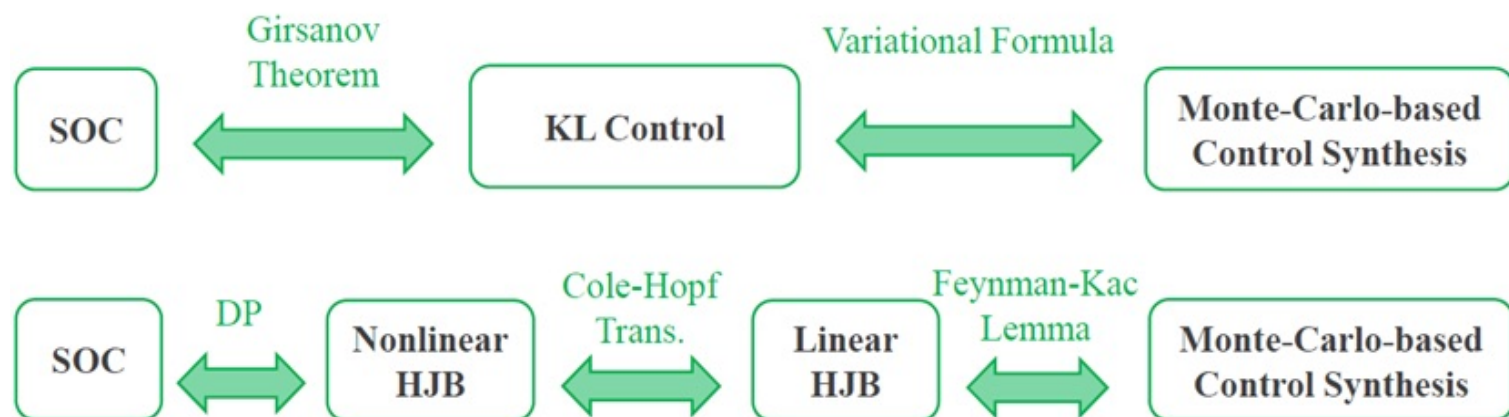


Figure 8.2: Derivation of the path integral control algorithm.

Continuous-Time KL control

$$\min_{\mathcal{Q}} \mathbb{E}^{\mathcal{Q}} \left[\int_0^T C_t(x_t) dt + C_T^{\text{exit}}(x_T) \right] + \alpha D(\mathcal{Q} \parallel \mathcal{R})$$

$\mathcal{Q}$: Distribution under the controlled model!

$\mathcal{R}$: Distribution under the uncontrolled model!

Free-energy representation of value function

By the variational formula, the value function $V_t(x)$ can be written as

Note: $\mathbb{E}^R[\cdot]$ can be computed by Monte-Carlo simulation of (2) over $[t, T]$. (8.20)

The variational formula also gives the optimal distribution

$$Q^*(B) = \frac{1}{Z} \int_B \exp\left(-\frac{1}{\alpha} C(x_{[t, T]})\right) dR \quad (8.21)$$

where B is an arbitrary subset of the sample path space $\mathcal{X}_{[t, T]} \times \mathcal{U}_{[t, T]}$.

Q: How to calculate the optimal control input μ_t in real time using Monte Carlo?

Path Integral Control Algorithm

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Step 1: Generate N independent sample paths by simulating the uncontrolled dynamics. Use Euler-Maruyama method.

$$\hat{x}_{z+\Delta t} = \hat{x}_z + f_z(\hat{x}_z)\Delta t + g_z(\hat{x}_z)\Delta \hat{W}_z$$
$$z = t, t+\Delta t, \dots, T-\Delta t$$

- Initial condition $\hat{x}_t = x$ (current state)
- $\Delta \hat{W}_z \sim N(0, \Delta t I)$

Step 2: For each sample $i=1, 2, \dots, N$, compute the desirability score

$$\hat{r}_t(i) = \exp \left(-\frac{1}{\alpha} \sum_{z=t}^{T-\Delta t} C(\hat{x}_z(i)) \Delta t - \frac{1}{\alpha} C_T^{\text{exit}}(\hat{x}_T(i)) \right)$$

Compute also $\hat{r}_r = \sum_{i=1}^N \hat{r}_t(i)$

Step 3: The optimal control input is approximated by

So far, we saw that PIC algorithm 14
can solve the KL control problems.

Q: How general is the KL control problem?

Consider the SOC in the standard form:

$$\begin{cases} \min_u \mathbb{E} \left[\int_0^T (C(x_t) + \frac{1}{2} u_t^T R(x_t) u_t) dt + C^{\text{exit}}(x_T) \right] \\ \text{s.t. } dx_t = (f_t(x_t) + g_t(x_t) u_t) dt + \sigma_t(x_t) dW_t. \end{cases} \quad (8.34)$$

Theorem 8.1 If there exists a
constant $\alpha > 0$ such that

"Matching condition"

then (8.34) can be reformulated as
a KL control problem.

$\Rightarrow$ PIC algorithm is applicable.

Deriving PIC algorithm using

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the Feynman-Kac formula

This is the original approach by Kappen.

- Recall that solving the SOC (8.34) is equivalent to solving the HJB PDE:

$$\begin{cases} -\partial_t V = -\frac{1}{2} (\partial_x V)^T g R^{-1} g^T (\partial_x V) + C \\ \quad + f^T \partial_x V + \frac{1}{2} \text{Tr}(\sigma \sigma^T \partial_x^2 V) \\ V(x, T) = \psi(x). \quad (\text{nonlinear}) \end{cases}$$

- Apply change of variables

$$V(x, t) = -\alpha \log Z(x, t)$$

- HJB becomes $(\Rightarrow \text{Exercise 8.1})$

$$\begin{cases} Z(x, T) = \exp\left(-\frac{\psi(x)}{\alpha}\right) \\ (\text{still nonlinear}) \end{cases}$$

- Under the matching condition, this PDE becomes

(8.52)

$$\begin{cases} -\partial_t Z = f_t^\top \partial_x Z - \frac{c}{\alpha} Z + \frac{1}{2} \text{Tr}(\sigma \sigma^\top \partial_x^2 Z) \\ Z(x, t) = \exp\left(-\frac{\psi(x)}{\alpha}\right) \end{cases} \quad (\text{linear!})$$

Theorem (Feynman - Kac formula)

The solution $Z(x, t)$ for the PDE (8.52) can be expressed as

where the expectation $\mathbb{E}$ is w.r.t the law of

$$dx_t = f_t dt + \sigma_t dw_t$$

- Hence, we obtained a result equivalent to (8.20).
- The same Monte-Carlo algorithm can be used.